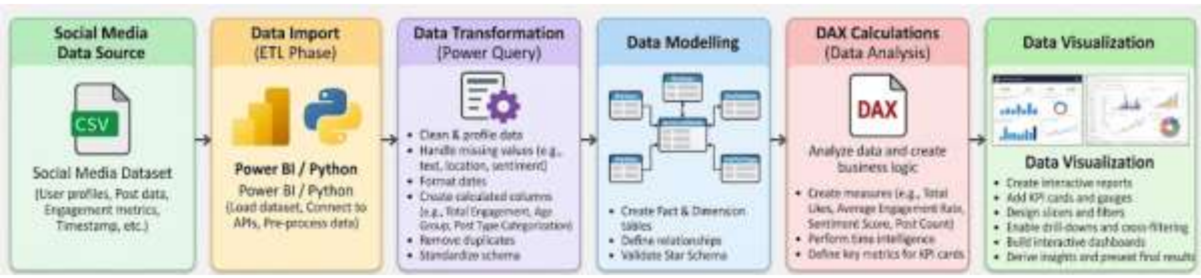
Ex.no: 8 MINI PROJECT – SOCIAL MEDIA DATA ANALYTICS

Aim: To develop an interactive Social Media Data Analytics Dashboard using Power BI and Tableau to analyze user posts, engagement metrics (likes, shares, comments), post types, user demographics, sentiment analysis, and platform performance.

Algorithm:

1. Import the Social Media Dataset into Power BI and Tableau.
2. Clean and transform the data using Power Query in Power BI and Data Source in Tableau.
3. Check and correct data types and remove unwanted or duplicate records.
4. Create required calculated fields such as Engagement Rate, Sentiment Category, and Account Age Group.
5. Create the required data model and star schema relationships in Power BI.
6. Create required DAX measures and KPIs in Power BI.
7. Create calculated fields and required measures in Tableau.
8. Create charts for post types, engagement levels, user demographics, sentiment breakdown, and reach.
9. Add cards, KPIs, slicers, filters, and interactive drill-downs in Power BI.
10. Add filters, interactive dashboards, and storytelling views in Tableau.
11. Design and format the dashboards in both Power BI and Tableau.
12. Analyze the visualizations and identify critical social media engagement trends.
13. Present the final results through interactive dashboards.

Architecture:



Program:

Python – Data Cleaning & Transformation

Python

```
import pandas as pd
from sqlalchemy import create_engine

# Load dataset
df = pd.read_csv("social_media_dataset.csv")

# Create a copy for processing
social_media = df.copy()

# Remove duplicate records
social_media.drop_duplicates(inplace=True)

# Handle missing values
social_media["post_text"] = social_media["post_text"].fillna("No Text")
social_media["platform"] = social_media["platform"].fillna("Unknown")
social_media["user_location"] = social_media["user_location"].fillna("Unknown")
social_media["sentiment"] = social_media["sentiment"].fillna("Neutral")
social_media["likes"] = social_media["likes"].fillna(0)
social_media["shares"] = social_media["shares"].fillna(0)
social_media["comments"] = social_media["comments"].fillna(0)

# Convert date columns
social_media["post_date"] = pd.to_datetime(social_media["post_date"])

# Create Total Engagement Column
social_media["total_engagement"] = (
    social_media["likes"] + social_media["shares"] + social_media["comments"]
)
```

```
# Create User Age Group
social_media["age_group"] = pd.cut(
    social_media["user_age"],
    bins=[0, 17, 24, 34, 50, 100],
    labels=["<18", "18–24", "25–34", "35–50", "50+"]
)

# Standardize column names
social_media.columns = (
    social_media.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)

# Display original and cleaned shape
print("Original Shape:", df.shape)
print("Cleaned Shape:", social_media.shape)
social_media.head()
```

MySQL – Database and Data Loading

SQL

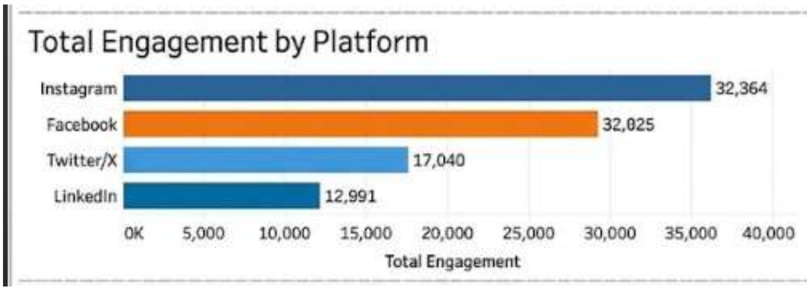
```
-- MySQL Command Prompt:
net start MySQL80
mysql -u root -p

-- MySQL:
CREATE DATABASE IF NOT EXISTS social_media_analytics;
```

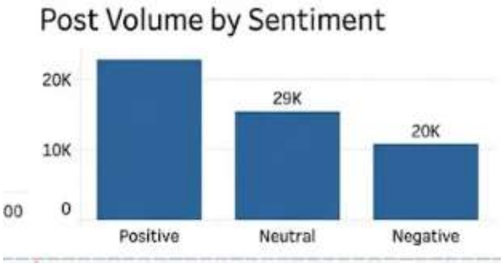
```
USE social_media_analytics;
```

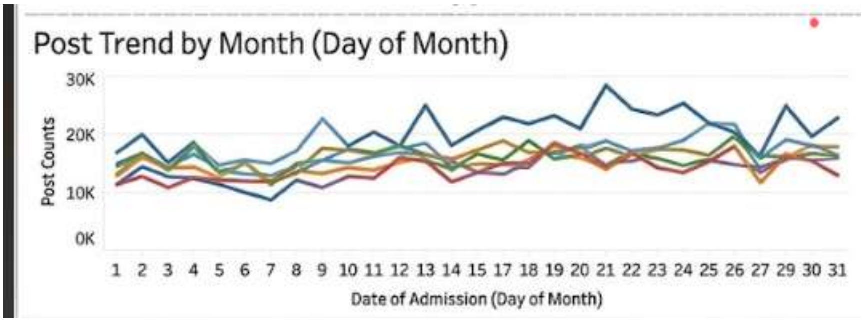
Python

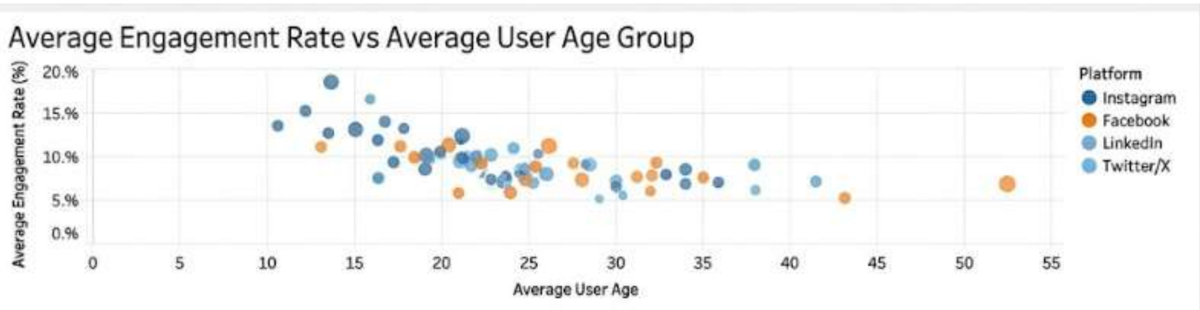
```python
# Python/Jupyter:
username = "root"
password = "YOUR_MYSQL_PASSWORD"
host = "localhost"
port = 3306

# Connect to MySQL database
db_engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/social_media_analytics"
)

# Load cleaned data into MySQL
social_media.to_sql(
    "social_media_data", con=db_engine, if_exists="replace", index=False, chunksize=5000
)
print("Data loaded successfully into MySQL.")

# Verify:
pd.read_sql(
    "SELECT COUNT(*) AS total_rows FROM social_media_data", db_engine
)
```

OUTPUT:









Total Posts

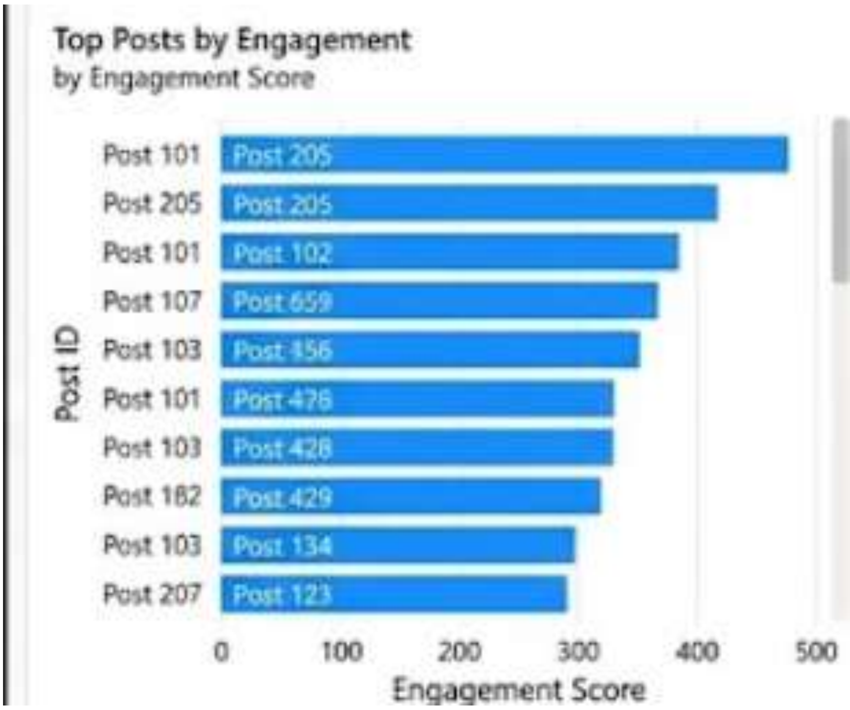
15K

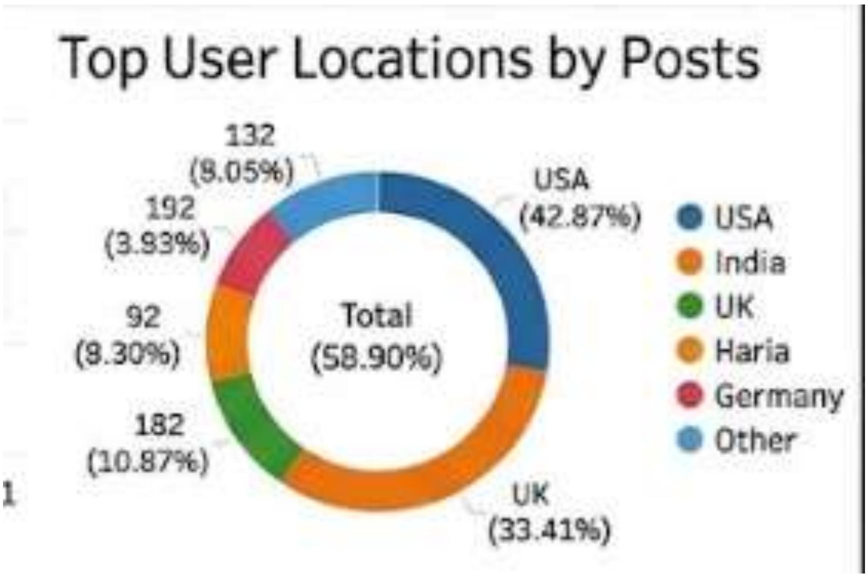
Total Likes

90K

Total Shares

40K





Result:

The interactive Social Media Data Analytics Dashboard was successfully built in both Power BI and Tableau. The system effectively imported, cleaned, and modeled raw post metrics, allowing end-to-end analysis of audience engagement, content performance trends, and sentiment distribution.